Worksheet 5.3 — Graphs of Sinusoidal Functions

Grade 12 Advanced Functions (MHF4U) · Unit 5: Trigonometric Functions

The Big Ideas

$y = a\sin(k(x-d)) + c$: amplitude $|a|$, period $\frac{2\pi}{k}$, phase shift d, midline $y = c$.

- Amplitude $= |a|$; midline $y = c$.
- Period $= \frac{2\pi}{k}$; phase shift $= d$.
- Read these four to sketch, or read the graph to write the equation.

Core Concepts

Sine and cosine in radians

Graphed over radians, $y = \sin x$ and $y = \cos x$ have period 2π, amplitude 1, and midline $y = 0$.

The transformed form

$y = a\sin(k(x-d)) + c$ gives amplitude $|a|$, period $\dfrac{2\pi}{|k|}$, phase shift d, and midline c.

Key points

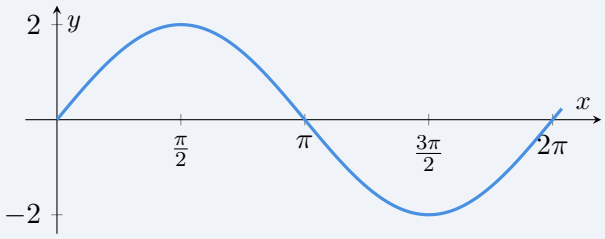
Plot the maxima, minima, and midline crossings for one period, then repeat to build the full wave.

Worked Examples

Example 1 — Amplitude

Amplitude of $y = 2\sin x$?

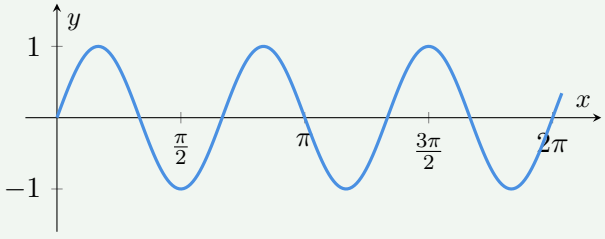
Solution. $|a| = 2$: the wave reaches ± 2:



Example 2 — Period

Period of $y = \sin(3x)$?

Solution. $\frac{2\pi}{3}$ three cycles in 2π:



Example 3 — Midline

Midline of $y = \cos x + 4$?

Solution. $c = 4$, so $y = 4$.

Example 4 — Longer period

Period of $y = \cos(\frac{x}{2})$?

Solution. $k = \frac{1}{2} \Rightarrow \frac{2\pi}{1/2} = 4\pi$.

Example 5 — Both

Amplitude and period of $y = 6\sin(2x)$?

Solution. $|a| = 6$; period $\frac{2\pi}{2} = \pi$.

Example 6 — Amplitude

Amplitude of $y = 8\cos x$?

Solution. 8.

Example 7 — Period

Period of $y = \cos(6x)$?

Solution. $\frac{2\pi}{6} = \frac{\pi}{3}$.

Example 8 — Midline

Midline of $y = \sin x - 6$?

Solution. $y = -6$.

Example 9 — Both

Amplitude and period of $y = 3\cos(4x)$?

Solution. Amplitude 3, period $\frac{\pi}{2}$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Amplitude of $y = 9\sin x$?

Question 2

Period of $y = \sin(8x)$?

Question 3

Midline of $y = \cos x - 5$?

Question 4

Period of $y = \sin(\frac{x}{4})$?

Question 5

Amplitude and period of $y = 5\cos(3x)$?

Question 6

Amplitude of $y = \frac{1}{2}\sin x$?

Question 7

Period of $y = \cos(3x)$?

Question 8

Midline of $y = \sin x + 5$?

Question 9

Amplitude and period of $y = 3\sin(\frac{x}{2})$?

Question 10

Period of $y = \sin x$?

Question 11

Amplitude of $y = 7\cos x$?

Question 12

Midline of $y = 2\sin x - 1$?

Question 13 — Challenge

State the amplitude, period and midline of $y = 4\sin(3x) - 2$.

Question Recap & Answer Key

Question Recap

1. Amplitude of $y = 9\sin x$?
2. Period of $y = \sin(8x)$?
3. Midline of $y = \cos x - 5$?
4. Period of $y = \sin(\frac{x}{4})$?
5. Amplitude and period of $y = 5\cos(3x)$?
6. Amplitude of $y = \frac{1}{2}\sin x$?
7. Period of $y = \cos(3x)$?
8. Midline of $y = \sin x + 5$?
9. Amplitude and period of $y = 3\sin(\frac{x}{2})$?
10. Period of $y = \sin x$?
11. Amplitude of $y = 7\cos x$?
12. Midline of $y = 2\sin x - 1$?
13. State the amplitude, period and midline of $y = 4\sin(3x) - 2$.

✓ Answer Key

1. 9
2. $\frac{\pi}{4}$
3. $y = -5$
4. 8π
5. amp 5, period $\frac{2\pi}{3}$
6. $\frac{1}{2}$
7. $\frac{2\pi}{3}$
8. $y = 5$
9. amp 3, period 4π
10. 2π
11. 7
12. $y = -1$
13. amp 4, period $\frac{2\pi}{3}$, midline $y = -2$